

Literature status: automorphic composition spectra on the little Bloch space and $VMOA$

Source arXiv:2207.08992; answer DOI:10.1007/s00009-017-0858-1

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Status: literature already answered. Both source questions have the same answer: for every parabolic or hyperbolic disk automorphism φ , the spectrum of C_φ is the full unit circle, on both $\mathcal{B}_0$ and $VMOA$.

Original questions

Allen, Le, and Pons ask at the end of Section 5 of their 2016 paper [1] (arXiv:2207.08992, PDF page 12):

Question 1. For φ a parabolic or hyperbolic automorphism, what is the spectrum of C_φ on the little Bloch space?

Question 2. For φ a parabolic or hyperbolic automorphism, what is the spectrum of C_φ on $VMOA$?

Separate answer paper

Matthew A. Pons's 2017 paper [2] explicitly cites [1], says in Section 4 that these are the remaining spectra to compute, and proves:

- **Theorem 4** (printed page 10 of 13): if φ is parabolic or hyperbolic, then

$$\sigma_{\mathcal{B}_0}(C_\varphi) = \partial\mathbb{D}.$$

- **Theorem 5** (printed pages 10–12): if φ is parabolic, then

$$\sigma_{VMOA}(C_\varphi) = \partial\mathbb{D}.$$

The proof constructs approximate eigenfunctions for every unimodular value.

- **Theorem 6** (printed page 12): if φ is hyperbolic, then

$$\sigma_{VMOA}(C_\varphi) = \partial\mathbb{D}.$$

Here Pons uses the canonical identification $VMOA^{**} = BMOA$, identifies the bidual of the restricted composition operator with C_φ on $BMOA$, and uses equality of the spectra of an operator and its bidual.

Thus the supporting author knew he was answering precisely the questions left by the source paper. No part of Questions 1 or 2 remains open for parabolic or hyperbolic disk automorphisms.

Search evidence

The run's exact-id/title indexes contained no existing packet. The decisive supporting paper was found through a Crossref exact-title query (DOI below), then checked in its full publisher PDF. A later arXiv paper on weighted Hardy spaces, arXiv:2302.08233, also cites both the 2016 Allen–Le–Pons paper and the surrounding automorphic-spectrum literature, but is not needed for the answer.

References

- [1] R. F. Allen, T. M. Le, and M. A. Pons, *Spectrum of a composition operator with automorphic symbol*, *Involve* 9 (2016), no. 5, 813–826; arXiv:2207.08992; DOI: 10.2140/involve.2016.9.813.
- [2] M. A. Pons, *The Spectrum of a Composition Operator with Automorphic Symbol*, *Mediterranean Journal of Mathematics* 14 (2017), article 53; DOI: 10.1007/s00009-017-0858-1.